KIT CARSON COUNTY AP, CO, USA

WMO: **724689**

Lat: **39.245N** Lon: **102.284W** Elev: **4192** StdP: **12.60** Time Zone: **-7.00 (NAM)** Period: **99-19** WBAN: **03026**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	12	0.7	6.0	-6.1	4.6	5.3	-1.5	5.9	14.0	34.4	32.4	30.4	35.0	10.6	340	0.691	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	28.1	97.3	64.9	94.4	64.5	91.4	64.3	69.5	84.9	68.5	84.3	67.5	83.7	12.8	170	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	65.5	110.3	74.4	64.0	104.6	72.8	63.0	100.8	72.1	36.2	85.0	35.2	84.0	34.2	83.4	74.5	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	31.3	26.9	24.1	DB	-7.9	101.5	4.8	2.3	-11.4	103.1	-14.2	104.5	-16.8	105.7	-20.3	107.4	(4)
(5)				WB	-8.5	72.1	4.7	1.5	-11.9	73.2	-14.7	74.2	-17.3	75.0	-20.7	76.1	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	51.6	30.4	31.4	41.2	49.3	58.7	70.3	75.9	73.1	65.1	51.5	40.0	30.5	(6)
(7)			DBStd	18.62	10.56	11.45	10.16	9.16	8.67	7.03	5.05	5.25	8.11	9.77	10.27	10.37	(7)
(8)			HDD50	2612	607	526	301	121	24	0	0	0	5	100	322	606	(8)
(9)			HDD65	5839	1072	942	736	474	230	31	3	7	98	426	749	1071	(9)
(10)			CDD50	3180	1	4	30	99	294	608	804	715	457	145	22	1	(10)
(11)			CDD65	931	0	0	0	2	35	189	342	257	100	6	0	0	(11)
(12)			CDH74	12523	0	5	45	180	772	2557	4236	2924	1494	295	15	0	(12)
(13)			CDH80	6037	0	1	5	36	273	1253	2316	1423	651	79	0	0	(13)
(14)	Wind		WSAvg	12.0	11.4	12.3	12.9	14.2	13.1	12.3	11.0	10.4	11.1	11.6	11.7	11.5	(14)
(15)	Precipitation		PrecAvg	15.5	0.3	0.4	0.9	1.3	2.9	2.4	2.2	1.9	1.3	0.8	0.5	0.4	(15)
(16)			PrecMax	27.8	1.8	2.8	4.9	3.4	4.6	7.9	6.4	6.5	4.5	3.0	1.6	1.5	(16)
(17)			PrecMin	8.5	0.0	0.0	0.0	0.0	0.4	0.3	0.5	0.0	0.0	0.0	0.0	0.0	(17)
(18)			PrecStd	4.1	0.4	0.6	1.1	0.8	1.2	1.6	1.3	1.5	1.2	0.8	0.5	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		66.3	71.8	79.4	84.1	91.4	100.6	100.2	98.5	95.4	86.8	76.3	64.3	(19)
(20)			MCWB		43.9	46.5	50.9	53.4	60.2	62.9	65.2	65.0	62.4	56.4	49.7	44.4	(20)
(21)		2%	DB		60.6	63.2	73.2	79.3	87.2	95.5	97.9	94.5	91.1	81.8	71.3	58.8	(21)
(22)			MCWB		42.1	43.6	48.3	52.2	58.6	63.3	65.4	64.7	61.6	54.9	48.2	41.6	(22)
(23)		5%	DB		53.8	56.8	67.5	74.4	82.2	91.7	95.3	91.4	88.0	76.8	64.6	53.3	(23)
(24)			MCWB		39.4	40.6	46.9	51.5	58.2	63.0	65.4	65.0	61.2	53.5	46.1	39.0	(24)
(25)		10%	DB		47.4	50.3	61.3	69.2	78.1	87.7	92.3	88.6	83.1	70.7	58.6	47.3	(25)
(26)			MCWB		36.5	37.8	44.5	50.5	56.9	63.1	65.8	65.4	60.5	51.6	43.3	36.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		45.8	48.1	54.4	57.9	65.1	69.8	71.9	70.9	67.3	60.7	52.2	45.2	(27)
(28)			MCDB		63.2	68.6	71.4	73.1	81.6	84.8	86.2	85.2	83.4	75.6	68.9	62.2	(28)
(29)		2%	WB		42.3	44.1	50.5	55.4	62.6	67.9	70.0	69.3	65.4	57.9	49.6	42.2	(29)
(30)			MCDB		58.3	61.2	67.5	70.8	77.8	83.6	85.4	84.5	81.1	72.5	67.4	57.2	(30)
(31)		5%	WB		39.8	41.3	48.0	53.3	60.6	66.3	68.9	68.3	64.0	55.4	47.1	39.4	(31)
(32)			MCDB		52.7	55.4	65.5	68.7	76.2	82.7	84.9	83.6	80.0	71.5	63.7	52.1	(32)
(33)		10%	WB		36.8	38.3	45.2	51.1	58.9	64.8	67.9	67.2	62.3	53.1	44.2	36.7	(33)
(34)			MCDB		47.0	50.0	60.1	67.0	73.4	81.4	84.5	82.4	77.1	67.9	57.9	47.0	(34)
(35)	Mean Daily Temperature Range		MDBR		26.2	25.7	28.7	28.0	27.7	28.3	28.1	26.7	28.5	28.3	28.0	26.1	(35)
(36)		5% DB	MCDBR		35.2	36.3	38.8	38.3	35.1	34.2	33.6	31.8	34.7	37.7	37.4	34.2	(36)
(37)			MCWBR		20.2	20.1	19.0	17.0	13.8	10.5	9.5	9.8	11.4	16.2	19.0	20.2	(37)
(38)		5% WB	MCDBR		33.0	34.3	35.5	32.0	30.2	29.6	27.9	26.7	28.9	31.5	35.4	32.5	(38)
(39)			MCWBR		19.8	19.8	18.4	16.4	13.5	11.3	9.8	9.9	10.8	15.4	18.4	19.8	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.245	0.254	0.280	0.303	0.329	0.348	0.364	0.352	0.321	0.283	0.255	0.246	(40)
(41)		taud			2.580	2.571	2.500	2.460	2.448	2.422	2.426	2.437	2.503	2.578	2.581	2.566	(41)
(42)		Ebn at Noon			304	314	313	309	301	294	288	290	294	298	297	295	(42)
(43)		Edh at Noon			22	26	31	34	36	36	36	35	30	25	22	21	(43)
(44)	All-Sky Solar Radiation	RadAvg			819	1074	1489	1829	2109	2363	2342	2062	1686	1228	896	716	(44)
(45)		RadStd			56	87	118	89	126	100	84	87	88	102	63	42	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page